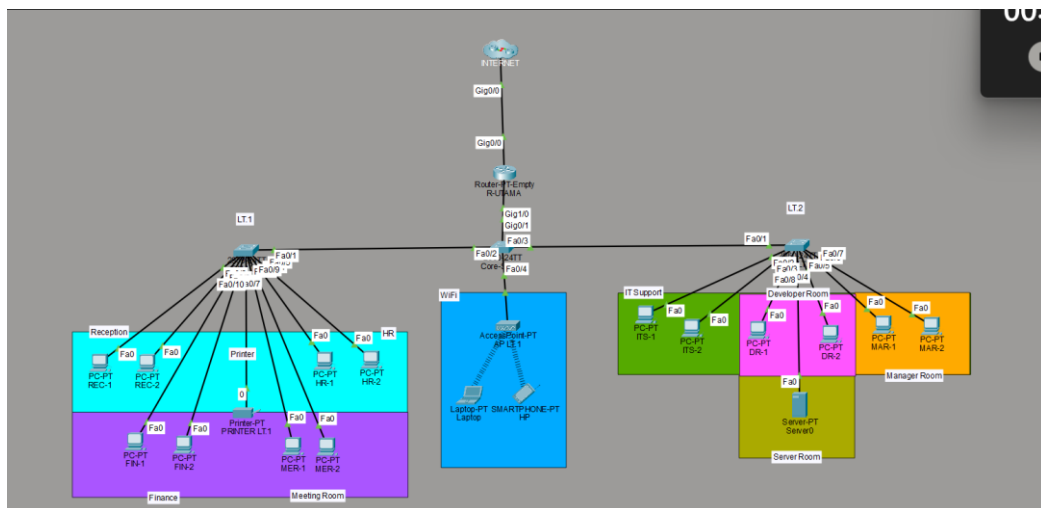


PT Adul Teknologi memiliki kantor dua lantai dengan total dua segmen jaringan utama. Lantai 1 digunakan untuk operasional administrasi, sedangkan lantai 2 digunakan untuk operasional teknis dan ruang server. Perusahaan menginginkan seluruh host memperoleh alamat IP secara otomatis, mampu mengakses layanan internal maupun Internet, serta dapat mengakses layanan menggunakan nama domain.

Topologi & Table Rancangan



Kebutuhan Perangkat Keras :

No	Perangkat	Jumlah
1	Router	1 Unit
2	Switch	3 Unit
3	Server	1 Unit
4	Access Point	1 Unit
5	PC Rception & Printer Area	2 Unit
6	PC HR	2 Unit
7	PC Finance	2 Unit
8	PC Meeting Room	2 Unit
9	Printer	1 Unit
10	PC IT Support	2 Unit
11	PC Developer Room	2 Unit
12	PC Manager Room	2 Unit

Rancangan VLAN :

VLAN ID	Nama VLAN	Network	Gateway	Digunakan Oleh
100	Network-LT.1	172.160.1.0/24	172.160.1.1	Seluruh perangkat LT.1
200	Network-LT.2	172.160.2.0/24	172.160.2.1	Seluruh perangkat LT.2
30	Wireless	172.160.3.0/24	172.160.3.1	Access Point & Wireless Client

Rancangan Port VLAN :

Device	Interface	Mode	VLAN	Keterangan
Switch LT1	Fa0/2–Fa0/10	Access	100	Host LT.1
Switch LT1	Fa0/1	Trunk	100,200,30	Ke Core Switch
Switch LT2	Fa0/2–Fa0/9	Access	200	Host LT.2
Switch LT2	Fa0/1	Trunk	100,200,30	Ke Core Switch
Core	Fa0/2	Trunk	100,200,30	Ke LT.1
Core	Fa0/3	Trunk	100,200,30	Ke LT.2
Core	Fa0/4	Access	30	Access Point
Core	Gi0/1	Trunk	100,200,30	Ke Router

WAN network plan :

Network	Device	Interface	IP Address
200.10.10.0/24	Router Utama	Gig0/0	200.10.10.2
200.10.10.0/24	Router Publik	Gig0/1	200.10.10.1
8.8.8.0/24	Router Publik	Gig1/0	8.8.8.1
8.8.8.0/24	Server Internet	Gig1	8.8.8.2

Server Network plan :

Server	IP Address	Fungsi
Server Lokal	172.160.2.8	DNS, Web Server

Server Internet	8.8.8.2	Web Server Publik
--------------------	---------	----------------------

Konfigurasi DHCP LT.1 – 2

1. Diketahui Network pada LT.1 adalah 172.160.1.0/24 dan network LT.2 adalah 172.160.2.0/24.
2. Konfigurasi pertama kali adalah membuat VLAN dari switch di setiap lantai. LT.1 Vlan 100, LT.2 vlan 200. Setiap port pada host di masukkan ke dalam vlan tersebut. Kecuali port fa0/1 di setiap switch karrena terhubung ke core switch.
3. Pada port fa0/1 di switch di buat mode trunk.
4. Pada core switch di buat trunk untuk port yang terhubung ke switch LT.1, switch LT.2 dan router.
5. Pada router di buat routing inter-vlan di port 1/0. Berarti ada 2 subinterfaces yaitu 1/0.100 dan 1/0.200. di buat dhcp pool juga agar setiap host mendapat ip secara otomatis.

Show vlan brief :

```
SW-LT1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	Raception	active	
20	HR	active	
30	Finance	active	
40	Meeting-Room	active	
100	Network-LT.1	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-LT1#
```

Copy Paste

```
SM-LT.2#show vlan br
SM-LT.2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
50	IT-Support	active	
60	Developer-Room	active	
70	Manager-Room	active	
80	Server-Room	active	
200	Network-LT.2	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SM-LT.2#
```

Show interfaces trunk :

```
SW-CORE#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/2	auto	n-802.1q	trunking	1
Fa0/3	on	802.1q	trunking	1
Gig0/1	on	802.1q	trunking	1

```
Port Vlan allowed on trunk
```

Fa0/2	100
Fa0/3	200
Gig0/1	100,200

```
Port Vlan allowed and active in management domain
```

Fa0/2	none
Fa0/3	none
Gig0/1	none

```
Port Vlan in spanning tree forwarding state and not pruned
```

Fa0/2	none
Fa0/3	none

```
SW-LT1#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

```
Port Vlan allowed on trunk
```

Fa0/1	100
-------	-----

```
Port Vlan allowed and active in management domain
```

Fa0/1	100
-------	-----

```
Port Vlan in spanning tree forwarding state and not pruned
```

Fa0/1	100
-------	-----

```
SM-LT.2#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
SM-LT.2#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

```
Port Vlan allowed on trunk
```

Fa0/1	200
-------	-----

```
Port Vlan allowed and active in management domain
```

Fa0/1	200
-------	-----

```
Port Vlan in spanning tree forwarding state and not pruned
```

Fa0/1	none
-------	------

Show ip interface brief :

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0       200.10.10.2     YES manual up      up
GigabitEthernet1/0       unassigned      YES unset  up      up
GigabitEthernet1/0.100   172.160.1.1     YES manual up      up
GigabitEthernet1/0.200   172.160.2.1     YES manual up      up
```

Show dhcp pool :

```
Router#show ip dhcp pool

Pool Network-LT.1 :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 0
  Excluded addresses              : 2
  Pending event                  : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  172.160.1.1        172.160.1.1      - 172.160.1.254      0 / 2 / 254

Pool Network-LT.2 :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 0
  Excluded addresses              : 2
  Pending event                  : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  172.160.2.1        172.160.2.1      - 172.160.2.254      0 / 2 / 254
Router#
```

Troubleshooting:

Permasalahan terjadi karena pada **Core Switch** belum dibuat VLAN yang sama dengan VLAN yang digunakan pada switch di setiap lantai (VLAN 100 dan VLAN 200). Akibatnya, meskipun port antar-switch telah dikonfigurasi sebagai **trunk**, frame yang membawa tag VLAN tersebut tidak dapat diteruskan oleh Core Switch sehingga permintaan **DHCP Discover** dari host tidak pernah mencapai router (DHCP Server).

Solusi:

Buat VLAN yang sesuai (VLAN 100 dan VLAN 200) pada Core Switch, kemudian pastikan seluruh port yang menghubungkan Core Switch dengan switch lantai dan router dikonfigurasi sebagai **trunk** serta mengizinkan VLAN yang diperlukan (switchport trunk allowed vlan 100,200). Setelah konfigurasi tersebut diterapkan, frame VLAN dapat diteruskan dengan benar sehingga host berhasil memperoleh alamat IP dari DHCP Server.

```

SW-CORE(config)#vlan 100
SW-CORE(config-vlan)#name Network-LT.1
SW-CORE(config-vlan)#ex
SW-CORE(config)#vlan 200
SW-CORE(config-vlan)#name Network-LT.2
SW-CORE(config-vlan)#ex
SW-CORE(config)#
SW-CORE(config)#ex
SW-CORE#
%SYS-5-CONFIG_I: Configured from console by console

```

```

SW-CORE#show vlan br
SW-CORE#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
100	Network-LT.1	active	
200	Network-LT.2	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

%SYS-5-CONFIG_I: Configured from console by console

```

```

show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/2	auto	n-802.1q	trunking	1
Fa0/3	on	802.1q	trunking	1
Gig0/1	on	802.1q	trunking	1


```

Port      Vlans allowed on trunk
Fa0/2     100,200
Fa0/3     100,200
Gig0/1    100,200

```



```

Port      Vlans allowed and active in management domain
Fa0/2     100,200
Fa0/3     100,200
Gig0/1    100,200

```



```

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/2     200
Fa0/3     200
Gig0/1    100,200

```

```

SW-CORE#

```

Konfigurasi AP

1. Membuat vlan baru di core switch dengan ID 30 nama AP. Memasukkan Port fa0/4 mode access ke vlan 30.

```
SW-CORE#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
30	AP	active	Fa0/4
100	Network-LT.1	active	
200	Network-LT.2	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	

2. Memasukkan vlan 30 ke mode trunk di port Gig0/1.

```
SW-CORE#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/2	auto	n-802.1q	trunking	1
Fa0/3	on	802.1q	trunking	1
Gig0/1	on	802.1q	trunking	1

```
Port Vlan allowed on trunk
```

Fa0/2	100,200
Fa0/3	100,200
Gig0/1	30,100,200

```
Port Vlan allowed and active in management domain
```

Fa0/2	100,200
Fa0/3	100,200
Gig0/1	30,100,200

```
Port Vlan in spanning tree forwarding state and not pruned
```

Fa0/2	100,200
Fa0/3	100,200
Gig0/1	30,200

3. Membuat sub interface 1/0.30 di router. Memasukkan IP 172.160.3.1 sebagai gateway

```
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	200.10.10.2	YES	manual	up	up
GigabitEthernet1/0	unassigned	YES	unset	up	up
GigabitEthernet1/0.30	172.160.3.1	YES	manual	up	up
GigabitEthernet1/0.100	172.160.1.1	YES	manual	up	up
GigabitEthernet1/0.200	172.160.2.1	YES	manual	up	up

4. Membuat IP pengecualian di dhcp pool nanti 172.160.3.1 & Membuat dhcp pool dengan nama AP

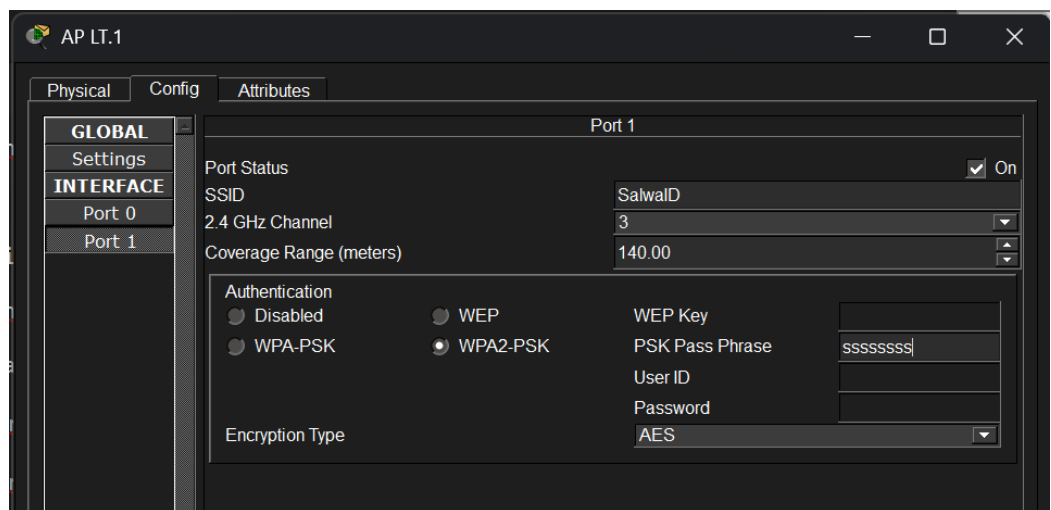
```

Pool AP :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 0
Excluded addresses                : 2
Pending event                     : none

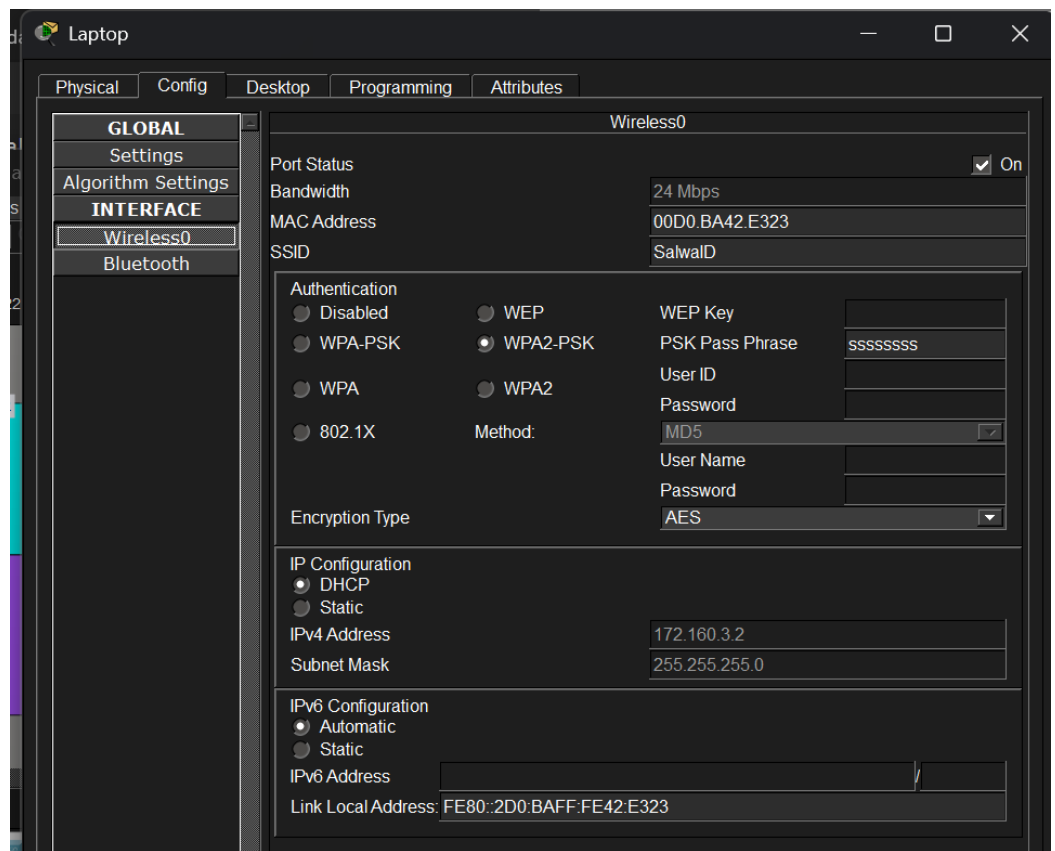
1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
172.160.3.1       172.160.3.1 - 172.160.3.254  0 / 2 / 254
Router#
Router##DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged 172.160.3.1.

```

5. Menyalakan Access Point. Mengatur jenis outentikasi ke WPA2-PSK. Mengatur SSID & Membuat password ssssssss. Dengan encryption type AES.



6. Menyambungkan ke perangkat host.

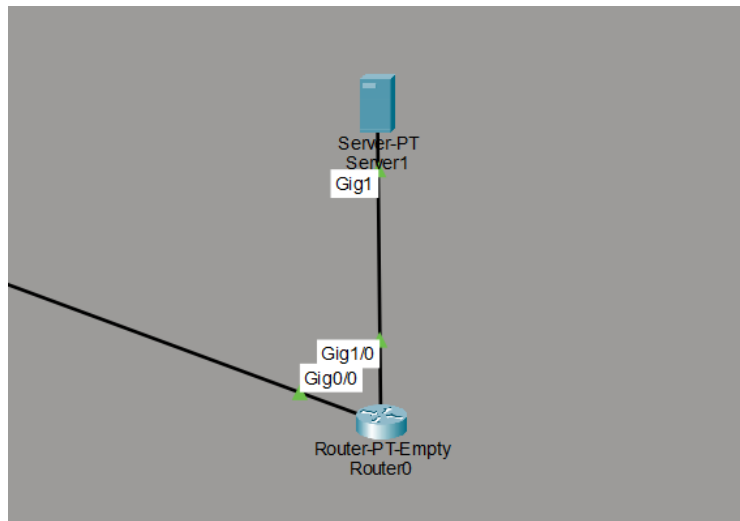


The screenshot shows the HP Network Assistant configuration window for the Wireless0 interface. The window is titled "HP" and has a menu bar with "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Config" tab is selected, and the "Wireless0" interface is chosen from the left sidebar. The main configuration area is divided into several sections:

- Wireless0**
 - Port Status: ☒ On
 - Bandwidth: 24 Mbps
 - MAC Address: 000C:CF1B:4965
 - SSID: SalwalID
- Authentication**
 - Disabled ☐ WEP ☐ WPA2-PSK ☒ WPA2 ☐ WPA ☐ 802.1X ☐
 - Method: MD5
 - WEP Key: [Empty field]
 - PSK Pass Phrase: ssssssss
 - User ID: [Empty field]
 - Password: [Empty field]
 - User Name: [Empty field]
 - Password: [Empty field]
- Encryption Type**: AES
- IP Configuration**
 - DHCP ☒ Static ☐
 - IPv4 Address: 172.160.3.3
 - Subnet Mask: 255.255.255.0
- IPv6 Configuration**
 - Automatic ☒ Static ☐
 - IPv6 Address: [Empty field]
 - Link Local Address: FE80::20C:CFFF:FE1B:4965

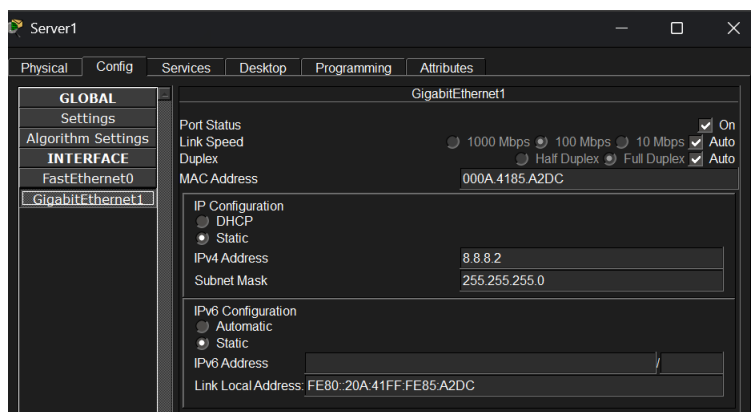
KONFIGURASI INTERNET

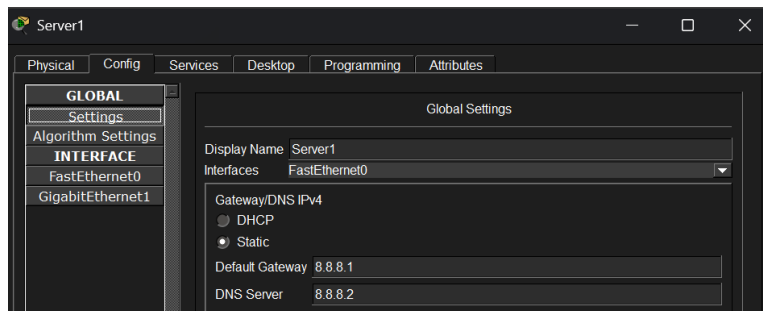
1. Terdapat perangkat server dan router di internet. Server sebagai penyedia peramban web serta dns server untuk jaringan internet misalnya. Sedangkan router bertindak sebagai gateway antara jaringan lokal menuju jaringan luar.



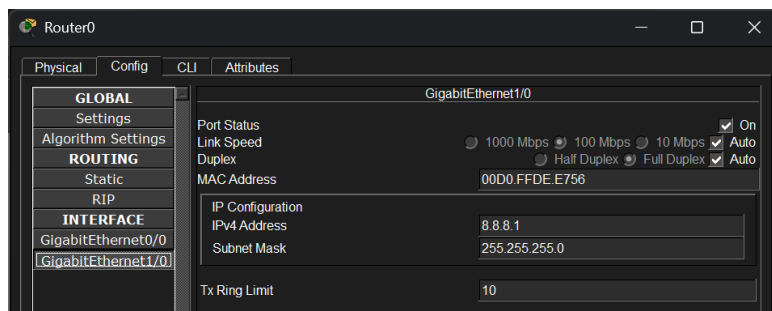
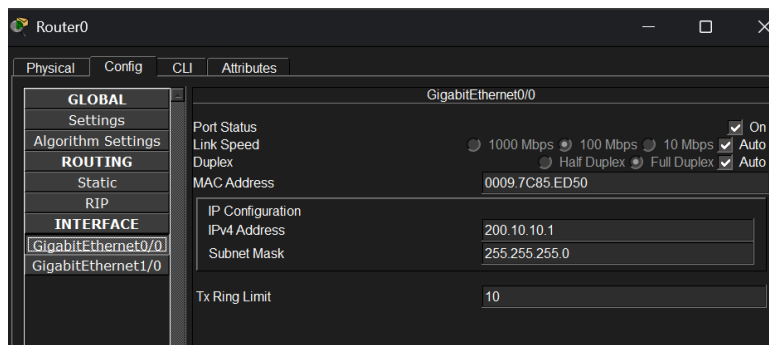
2. Langkah pertama adalah memberi IP address pada port router internet yang terhubung ke router utama dengan network 200.10.10.0/24. Pada port server yang terhubung ke router di beri ip dengan network 8.8.8.0/24. Setting pula default gateway dan dns servernya.

Server internet :

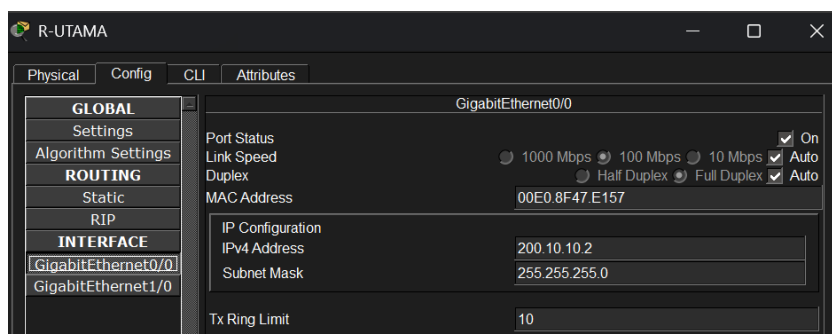




Router Internet :



Router Utama :



3. Melakukan routing menggunakan dynamic RIP routing. Pastikan anda memasukkan subnet yang terhubung pada setiap router agar router berhasil berkoumnikasi antara satu subnet dengan subnet yang lain.

```

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

      8.0.0.0/24 is subnetted, 1 subnets
C       8.8.8.0 is directly connected, GigabitEthernet1/0
R    172.160.0.0/16 [120/1] via 200.10.10.2, 00:00:06, GigabitEthernet0/0
C    200.10.10.0/24 is directly connected, GigabitEthernet0/0

```

```

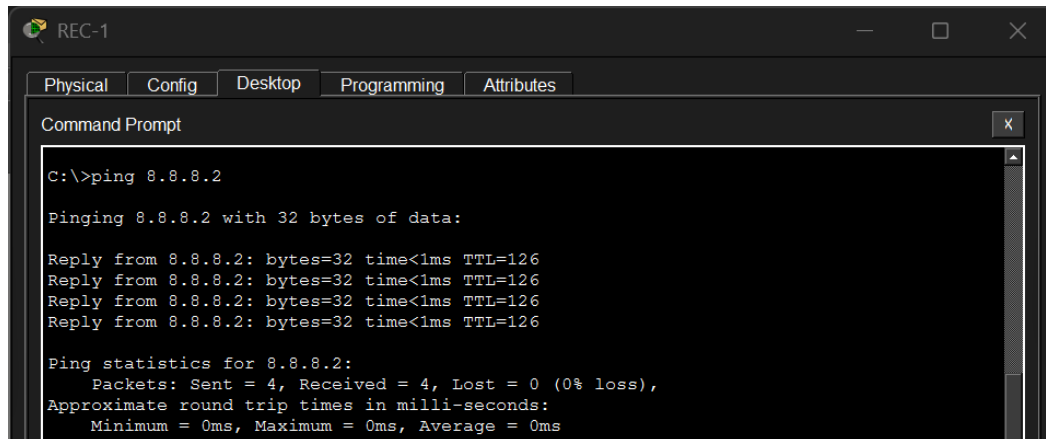
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

      8.0.0.0/24 is subnetted, 1 subnets
C       8.8.8.0 is directly connected, GigabitEthernet1/0
R    172.160.0.0/16 [120/1] via 200.10.10.2, 00:00:06, GigabitEthernet0/0
C    200.10.10.0/24 is directly connected, GigabitEthernet0/0

```

4. Tes ping ke alamat web server.



The screenshot shows a Windows Command Prompt window titled "REC-1". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Command Prompt" tab is active. The command prompt shows the following output:

```

C:\>ping 8.8.8.2

Pinging 8.8.8.2 with 32 bytes of data:

Reply from 8.8.8.2: bytes=32 time<1ms TTL=126
Reply from 8.8.8.2: bytes=32 time<1ms TTL=126
Reply from 8.8.8.2: bytes=32 time<1ms TTL=126
Reply from 8.8.8.2: bytes=32 time<1ms TTL=126

Ping statistics for 8.8.8.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

KONFIGURASI SERVER

1. Konfigurasi Server Internet

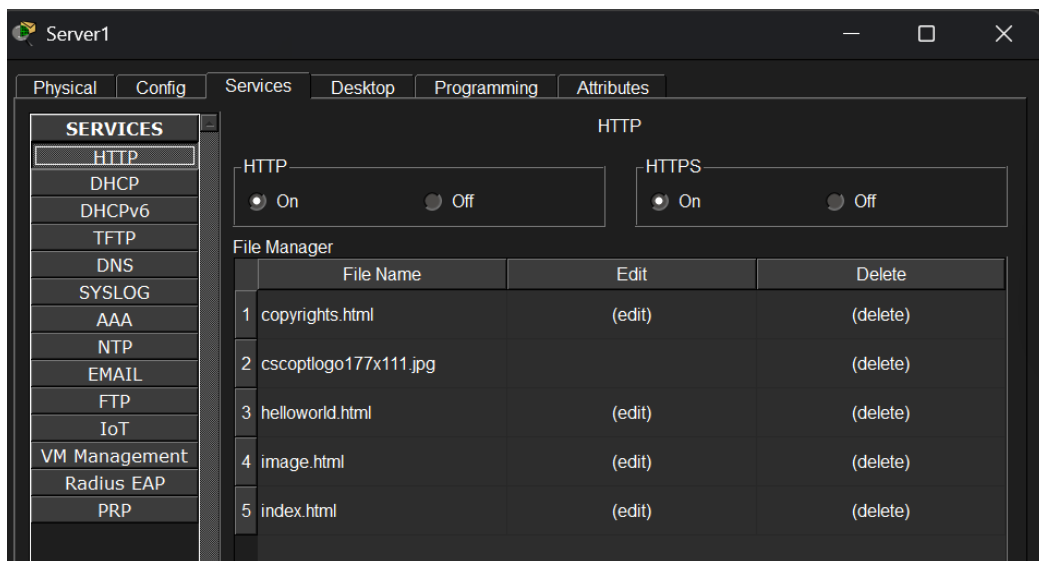
Server Internet berfungsi sebagai simulasi layanan yang berada di luar jaringan lokal perusahaan. Pada server ini aktifkan layanan **HTTP** dan **DNS**.

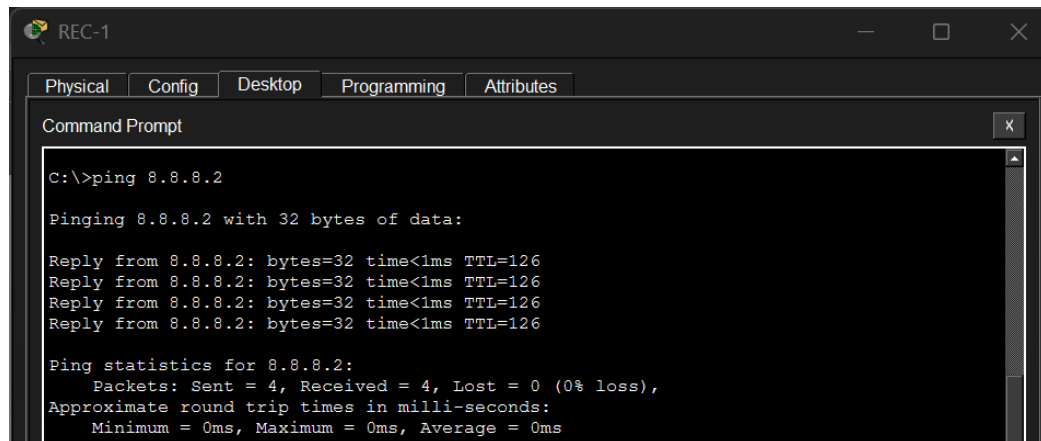
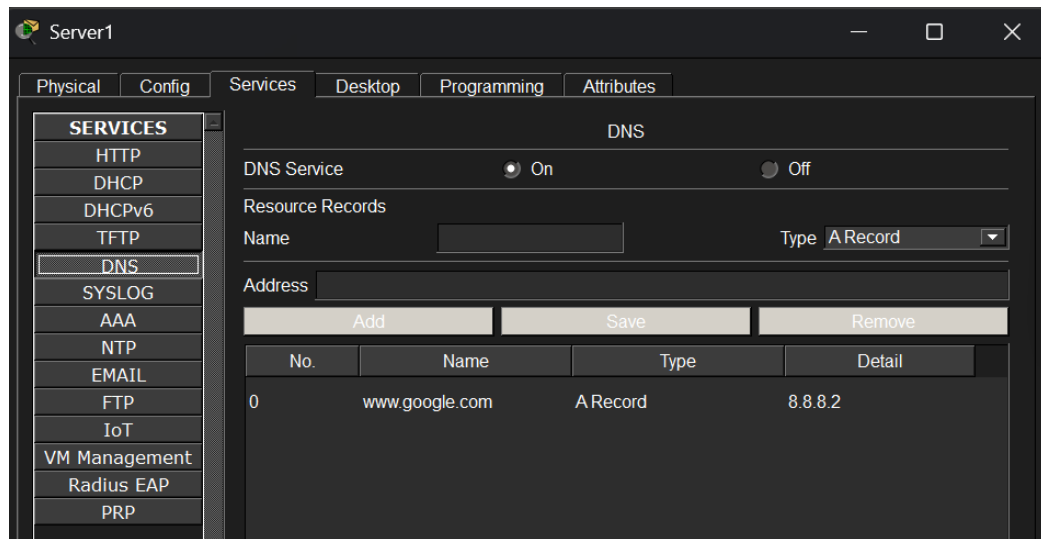
Karena pada simulasi Cisco Packet Tracer layanan web dan DNS ditempatkan pada perangkat yang sama, maka alamat IP Server Internet digunakan sebagai alamat DNS sekaligus alamat Web Server.

Lakukan konfigurasi DNS dengan menambahkan record berikut.

Domain	IP Address
www.google.com	8.8.8.2

Selanjutnya lakukan pengujian dengan melakukan **ping** ke alamat IP Server Internet untuk memastikan konektivitas jaringan telah berhasil.





2. Konfigurasi Server Lokal

Server Lokal berfungsi sebagai **DNS Server utama** bagi seluruh host di jaringan internal serta sebagai **Web Server** untuk layanan internal perusahaan.

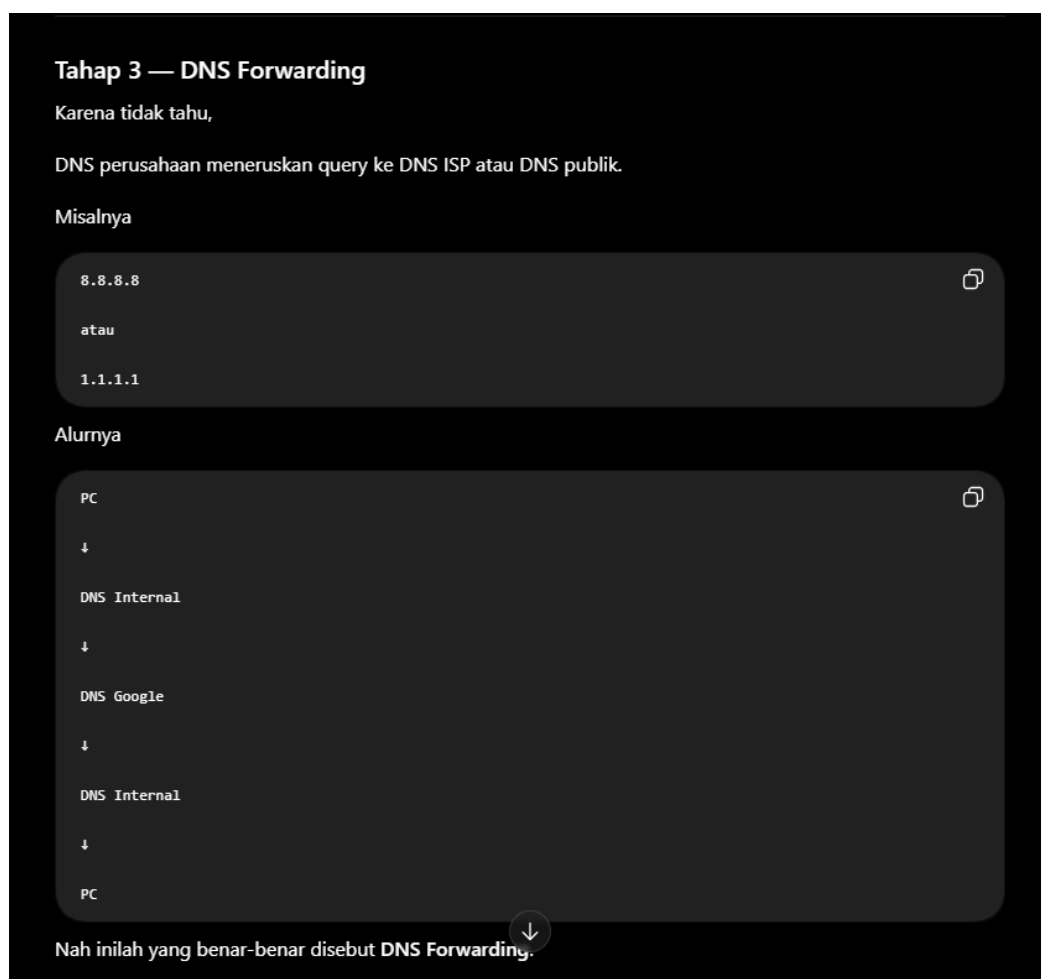
Lakukan konfigurasi berikut.

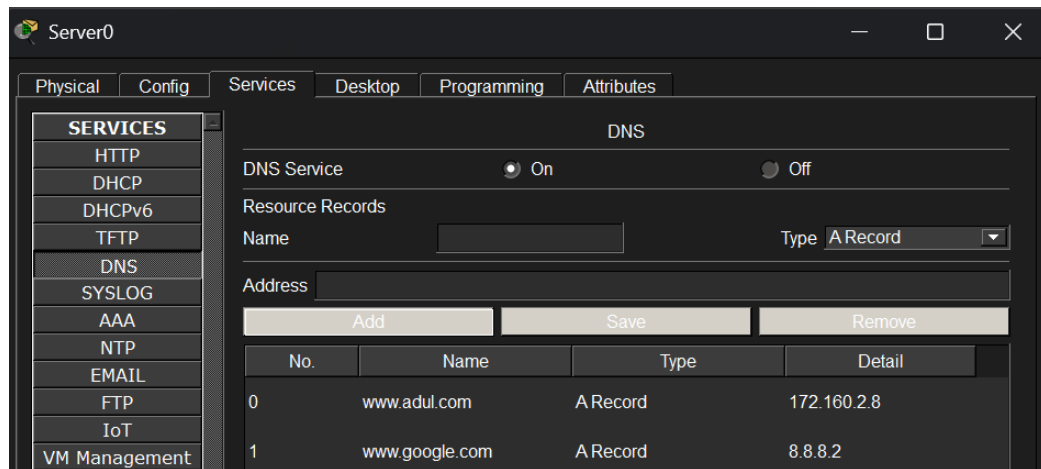
- Gunakan alamat IP **statis** agar alamat server tidak berubah.
- Aktifkan layanan **HTTP** untuk menyediakan layanan web internal.
- Aktifkan layanan **DNS**.

Tambahkan record DNS berikut.

Domain	IP Address	Fungsi
www.adul.com	172.160.2.8	Mengakses layanan web internal
www.google.com	8.8.8.2	Mengarahkan akses ke Server Internet

Catatan: Penambahan record www.google.com → 8.8.8.2 dilakukan secara **static mapping** karena DNS Server bawaan Cisco Packet Tracer tidak mendukung fitur **DNS Forwarding**. Pada implementasi jaringan nyata, DNS Server lokal umumnya akan meneruskan (forward) permintaan yang tidak dikenali ke DNS Server publik atau DNS milik ISP.





3. Konfigurasi DHCP

Pada konfigurasi DHCP Pool, parameter **DNS Server** diarahkan ke **Server Lokal**, bukan langsung ke Server Internet.

Contoh konfigurasi:

```
dns-server 172.160.2.8
```

Dengan konfigurasi tersebut seluruh host akan menggunakan Server Lokal sebagai DNS utama.

4. Alur Kerja Resolusi Nama Domain

Ketika pengguna mengakses domain www.google.com, host terlebih dahulu mengirimkan permintaan **DNS Query** ke DNS Server Lokal (172.160.2.8) yang diperoleh melalui DHCP.

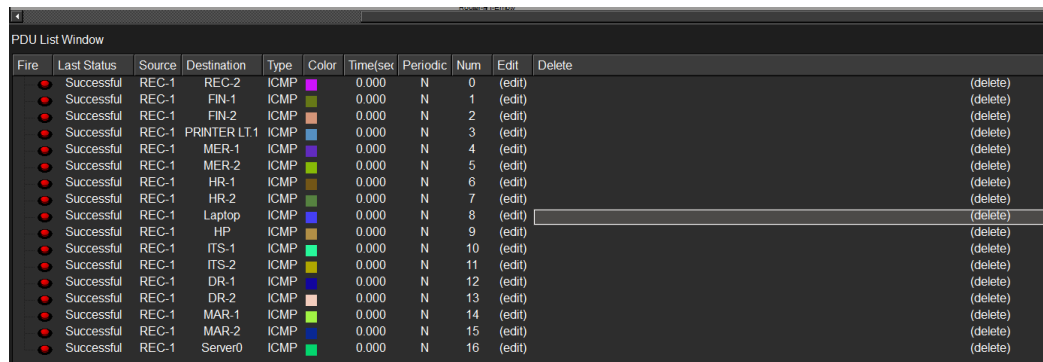
DNS Server Lokal kemudian memeriksa database DNS yang dimilikinya. Karena telah terdapat record statis www.google.com → **8.8.8.2**, DNS Server langsung mengembalikan alamat IP **8.8.8.2** kepada host tanpa melakukan proses DNS Forwarding.

Setelah memperoleh alamat IP tujuan, host mengirimkan paket data menuju **default gateway**. Paket tersebut diteruskan oleh Router Lokal menuju Router Internet melalui jaringan publik **200.10.10.0/24**, kemudian diteruskan ke Server Internet dengan alamat **8.8.8.2**.

Server Internet memproses permintaan tersebut dan mengirimkan balasan melalui jalur yang sama hingga akhirnya diterima kembali oleh host.

Tahap Pengujian

Pengujian di di wakili oleh device lantai 1. Mengeping seluruh perangkat host dalam jaringan termasuk access point dan jaringan lantai 2.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	REC-1	REC-2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	REC-1	FIN-1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	REC-1	FIN-2	ICMP		0.000	N	2	(edit)	(delete)
	Successful	REC-1	PRINTER LT.1	ICMP		0.000	N	3	(edit)	(delete)
	Successful	REC-1	MER-1	ICMP		0.000	N	4	(edit)	(delete)
	Successful	REC-1	MER-2	ICMP		0.000	N	5	(edit)	(delete)
	Successful	REC-1	HR-1	ICMP		0.000	N	6	(edit)	(delete)
	Successful	REC-1	HR-2	ICMP		0.000	N	7	(edit)	(delete)
	Successful	REC-1	Laptop	ICMP		0.000	N	8	(edit)	(delete)
	Successful	REC-1	HP	ICMP		0.000	N	9	(edit)	(delete)
	Successful	REC-1	ITS-1	ICMP		0.000	N	10	(edit)	(delete)
	Successful	REC-1	ITS-2	ICMP		0.000	N	11	(edit)	(delete)
	Successful	REC-1	DR-1	ICMP		0.000	N	12	(edit)	(delete)
	Successful	REC-1	DR-2	ICMP		0.000	N	13	(edit)	(delete)
	Successful	REC-1	MAR-1	ICMP		0.000	N	14	(edit)	(delete)
	Successful	REC-1	MAR-2	ICMP		0.000	N	15	(edit)	(delete)
	Successful	REC-1	Server0	ICMP		0.000	N	16	(edit)	(delete)

Setiap LAN juga berhasil mengakses gateway nya masing masing :



Successful	REC-1	R-UTAMA	ICMP		0.000	N	17	(edit)	(delete)
Successful	ITS-1	R-UTAMA	ICMP		0.000	N	18	(edit)	(delete)
Successful	Laptop	R-UTAMA	ICMP		0.000	N	19	(edit)	(delete)

Setiap LAN dapat mengakses domain server lokal dan internet :

